C1. Equal Multisets (Easy Version)

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: binary search, brute force, greedy, two pointers | https://codeforces.com/problemset/problem/2211/C1

This is the easy version of the problem. The difference between the versions is that in this version, the array *a* is guaranteed to be a permutation. You can hack only if you solved all versions of this problem.

Given an array a of size n, and a parameter k, an array b is called *cool* if the following conditions hold:

- For each i from k to n, the array $[a_{i-k+1},a_{i-k+2},\ldots,a_i]$ is a rearrangement of $[b_{i-k+1},b_{i-k+2},\ldots,b_i]$.

You are given two arrays a and b of size n, and an integer k. The array a **is guaranteed to be a permutation**[]. The array b only contains integers from 1 to n, and -1.

Determine if it is possible to replace all -1 in b with an integer from 1 to n, such that b is *cool* with respect to k.

[]A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array), and [1,3,4] is also not a permutation (n=3 but there is 4 in the array).

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and k ($1 \le k \le n \le 2 \cdot 10^5$).

The second line of each test case contains n integers $a_1,a_2,\ldots,a_n$ ($1 \le a_i \le n$). **In this version, it is guaranteed that each number from 1 to n appears exactly once.**

The third line of each test case contains n integers $b_1,b_2,\ldots,b_n$ ($1 \le b_i \le n$ or b_i=-1).

It is guaranteed that the sum of n across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output YES if possible, and NO otherwise. You may print each letter in either uppercase or lowercase.

Example

Input

```
4
5 5
1 2 3 4 5
3 1 5 2 4
5 4
4 1 2 5 3
2 -1 -1 -1 -1
6 4
1 2 4 3 5 6
-1 -1 3 -1 -1 -1
6 4
1 2 4 3 5 6
-1 -1 3 3 -1 -1
```

Output

```
YES
NO
YES
NO
```

Note

In the first test case, we have k=5. The only subarray of size 5 is [1,5]. We can see that b is a rearrangement of a, so the answer is YES.

In the second test case, it can be shown that it is impossible to replace all -1 in b such that every subarray of size k=4 are rearrangements of each other.